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1 Gulf of Mexico Shoreline Reconciliation - Methods & Results (RC1)

Author: Nate Murry - NOAA / NOS / CO-OPS **Prepared for:** VDatum Modeling Team, for use in SMS Modeling Software **Status:** Release Candidate 1 (RC1) - Modeler Review **Date:** 2026-08-18 **Processing platform:** Ubuntu 24.04.4 LTS, PostgreSQL 17 / PostGIS 3.6, GDAL/OGR

1.1 Executive Summary

We were provided several overlapping, high-resolution shoreline datasets covering adjacent regions of the Gulf of Mexico. In the overlap strips, the *same* physical features (islands, spits, marsh) were digitized independently by each dataset and do **not** coincide - e.g., the two halves of an island are offset and overlap in the middle without meeting. Overlapping, non-coincident linework of this kind breaks mesh generation in SMS.

This effort produced two reconciled, non-overlapping deliverables covering the full domain:

- **A merged shoreline LINE layer** - one coherent shoreline, redundant overlap removed, maximum source detail preserved.
- **A companion POLYGON layer** - closed land/island/water-body polygons derived from the line network, for defining land-vs-water regions in the mesh.

Both are delivered in WGS84 (EPSG:4326). All processing was **rule-based and quantified** - we removed redundant/duplicate linework and derived polygons, but we did **not** smooth, thin, or fabricate shoreline. Full source vertex detail is retained.

Validation performed (Section 6) indicates the products are structurally clean and should ingest into SMS without the overlap/topology failures that motivated this work: zero measured overlap between reconciled layers, 100% valid polygon geometries, and a quantified accounting of open (non-polygonizing) shoreline that is preserved in the line layer by design rather than falsely closed.

1.2 The Problem

Inputs (5 shoreline datasets):

Working color	Dataset (region)	Features	Native CRS
Purple	1a2206 UTM15 (NGS)	163,010	NAD83(2011)
Green	1a2206 UTM16 (NGS)	41,889	NAD83(2011)

Working color	Dataset (region)	Features	Native CRS
Orange	la2205 (NGS)	20,720	NAD83
Red	la2207 (NGS)	787	NAD83(2011)
(ref)	MS_TX (Notre Dame)	37,557	NAD83

Full source filenames are `la2206_cm_c_utm15_merge_shrln_class`, `la2206_cm_c_utm16_merge_shrln_class`, `la2205_cm_c_merge_shrln_class`, `la2207_cm_c_merge_shrln_class`, and `MS_TX_New_Shoreline`.

The four NGS tiles are the subject of this reconciliation. The Notre Dame MS_TX layer is a separate, coarser dataset used only for reference (Section 3) and is **not** modified or merged here.

Core issue: where adjacent NGS tiles overlap, they redundantly trace the same coastline with vertex-level disagreement. Merging them naively yields: - **Overlapping / crossing linework** (two tracings of one feature), and - In complex marsh, a “**tangle**” of thousands of criss-crossing segments and spurious micro-polygons.

Either condition is fatal to SMS mesh generation (unresolvable boundaries; sub-meter elements forcing crippling CFL timesteps).

1.3 Source Detail Assessment (why the NGS tiles, and why detail matters)

We quantified shoreline detail objectively as **vertices per kilometer** and average inter-vertex segment length (computed directly from geometry, no subjective judgment):

Dataset	Vertices / km	Avg segment length
Green (NGS)	710	1.4 m
Purple (NGS)	542	1.9 m
Orange (NGS)	493	2.0 m
MS_TX (Notre Dame)	111	9.0 m

The NGS tiles carry **4.5-6.4× the vertex density** of the Notre Dame layer - they trace the coastline in far finer detail. Preserving that detail is the primary goal; it is what makes this reconciliation worth doing rather than simply using the coarser existing product.

1.4 Method Evolution (what we tried, and why we changed)

The final method was reached by ruling out simpler approaches on evidence. Documenting this path is deliberate - it justifies the final design.

1.4.1 Attempt 1 - Concave-hull clipping (abandoned)

Initial approach (previously prototyped in ArcGIS/QGIS): convert one shoreline to points, build a `ST_ConcaveHull` “shrink-wrap” polygon, and use it to clip the overlapping neighbor. We reproduced this in PostGIS (grid-tiled for scalability) and tuned the concavity coefficient from 0.1 (loose) down to 0.001 (tight).

Why it failed - an inherent ceiling, not a tuning problem: a hull is a polygon *approximation* of where a shoreline is. No single concavity both hugs the shoreline tightly *and* sweeps up all overlapping neighbor linework. Loose hulls over-cut (delete good, unique data); tight hulls under-cut (leave residual overlap). At **every** concavity tested, measurable overlap remained. Because even one residual overlap breaks the model, the concave-hull method cannot meet the requirement regardless of tuning.

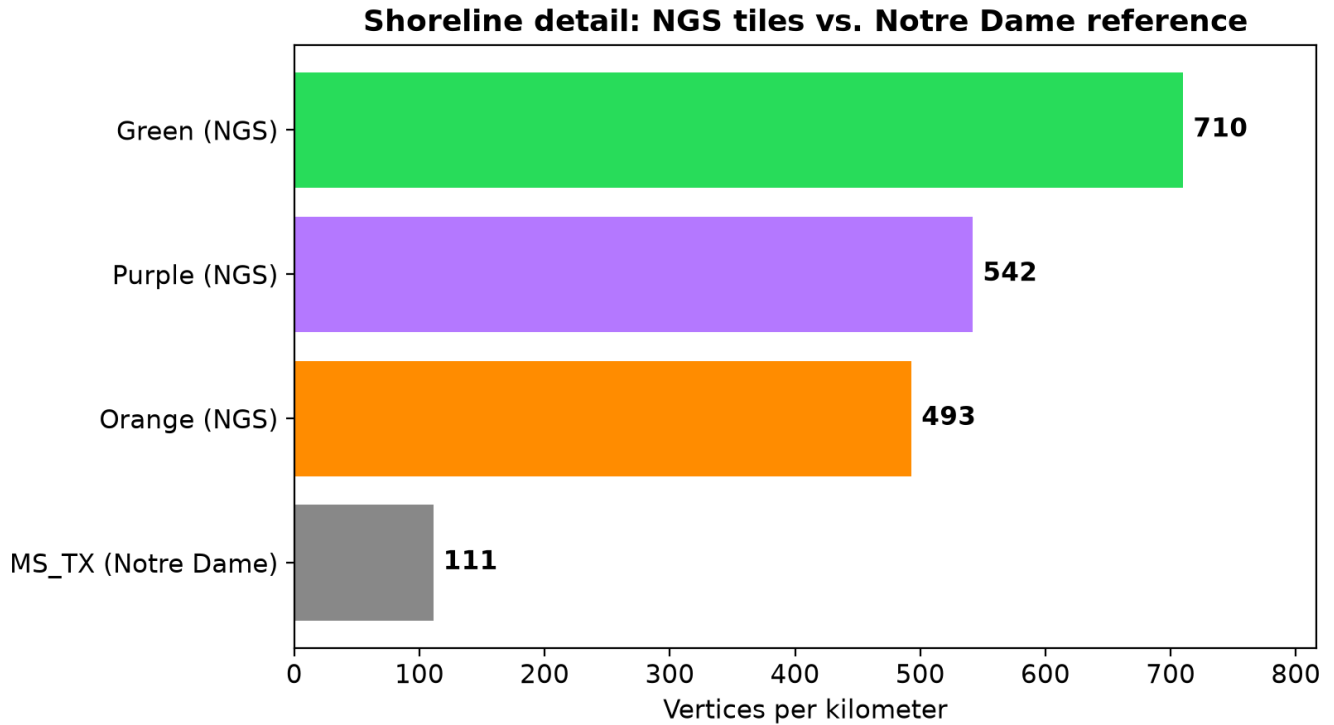


Figure 1: Shoreline detail: NGS tiles carry 4.5-6.4x the vertex density of the Notre Dame reference.

1.4.2 Attempt 2 - Buffer-difference overlap removal (adopted)

Replace the hull with a direct, physical rule:

Remove the portion of one shoreline that runs within N meters of another - i.e., where they trace the same feature - using `ST_Difference(lineA, ST_Buffer(lineB, N))`.

The parameter N is physically meaningful (“how close counts as duplicate”). Measured result: **0.0 m residual overlap** at both $N = 2$ m and $N = 5$ m - versus the hull, which always left overlap. Adopted at **$N = 5$ m**.

1.4.3 Fragment cleanup (buffer-difference debris)

Differencing shatters co-tracing zones into many short slivers. A length histogram showed thousands of sub-5 m fragments comprising ~2% of length, while genuine shoreline lived in long segments. Critically, **every leftover stub sat at exactly the buffer distance from the reference** - so *distance to the reference* is the discriminator.

We remove a fragment only if it is **both short (< 15 m) AND near the reference (< 6 m)**. This surgically removes clip debris while preserving genuine short features (a real short segment far from the reference, or a long segment near it, is kept). Result: ~1,000 debris fragments removed for < 1% length loss; all real segments retained.

1.4.4 Detail-based authority (the key refinement)

A simple “green always wins” precedence, applied during buffer-difference, revealed a problem in complex marsh: it could delete the *more detailed* tracing in favor of the *coarser* one, and where both datasets densely traced the same marsh (differing vertex-by-vertex), the merge produced a tangle of crossings.

We measured detail cell-by-cell (500 m grid) across the overlap zone. Neither dataset is uniformly finer: **green is denser in 289 cells, orange in 252**. The correct rule is therefore not a fixed precedence but **per-zone detail preference**:

In each cell, keep whichever dataset is more detailed; remove the other dataset’s *duplicate* linework (within 5 m), while preserving its *unique* (non-duplicated) detail.

Fig 4. Buffer-difference overlap removal at a seam

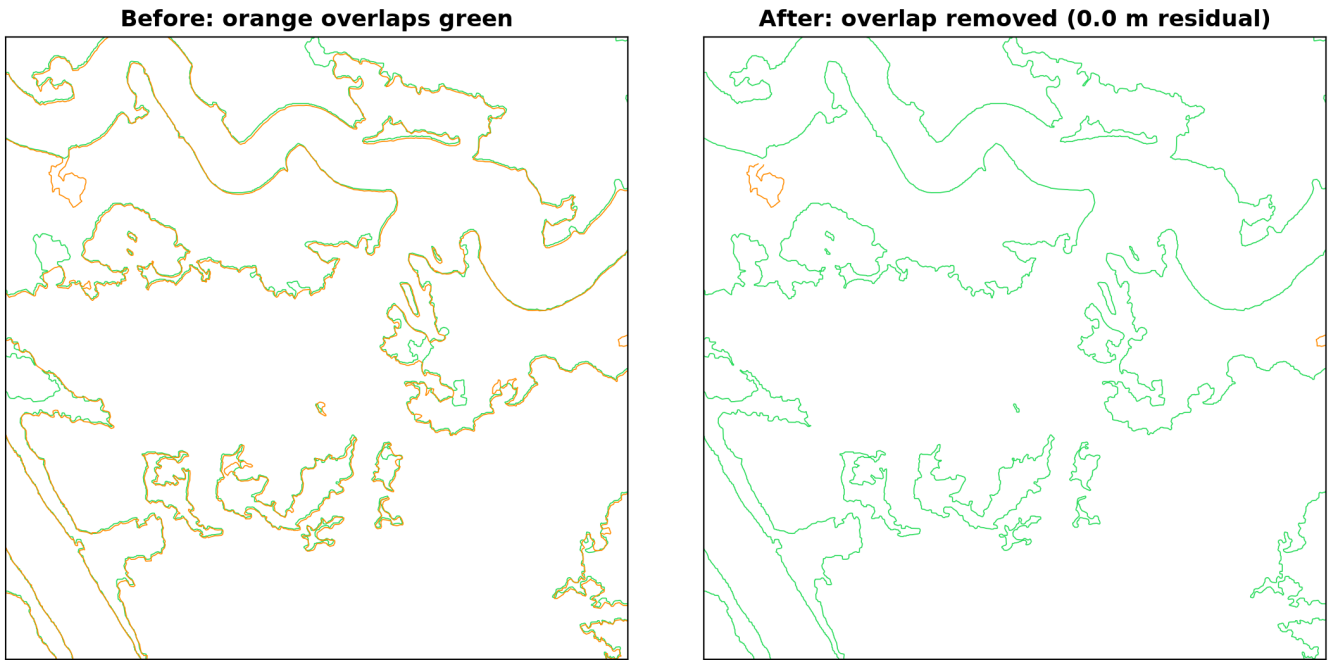


Figure 2: Buffer-difference overlap removal at a seam: orange linework within 5 m of green is erased (0.0 m residual overlap).

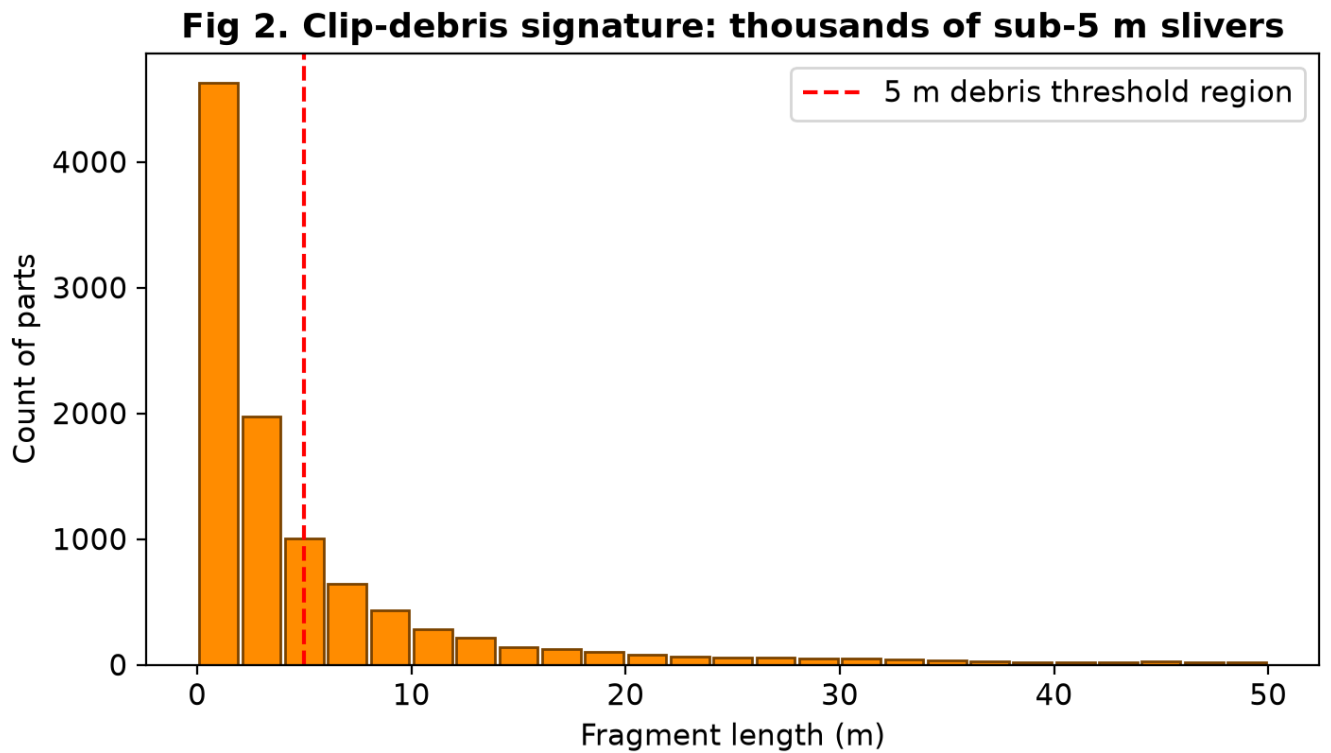
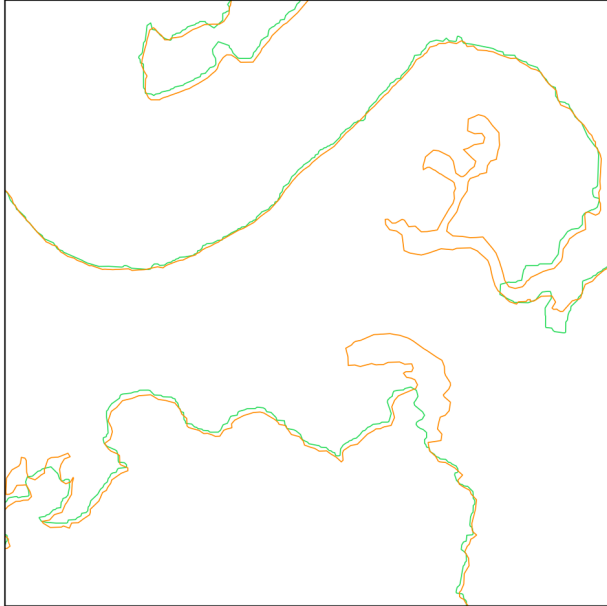


Figure 3: Clip-debris signature: thousands of sub-5 m fragments are removed; genuine shoreline lives in long segments.

This eliminates the tangle (a representative marsh location dropped from 1,076 green/orange crossings to zero) while preserving the finest available detail everywhere. Green and orange outside the overlap zone, and purple and red (which have negligible / no overlap), are carried through unchanged.

Fig 3. Marsh tangle resolved by detail-based authority

**Before: green + orange overlaid
(1,076 crossings — tangle)**



**After: detail-based reconciliation
(single coherent shoreline)**



Figure 4: Marsh tangle resolved: overlaid green+orange (1,076 crossings) becomes one coherent reconciled shoreline.

1.4.5 Polygon derivation and feature-size filtering

The reconciled line network is noded and polygonized (`ST_Node` then `ST_Polygonize`) to derive closed land/island/water polygons. Two cleanup passes follow:

- **Artifact slivers:** near-zero-area, needle-thin loops from noding are removed (identified by both tiny area *and* extreme thinness = $\text{perimeter} / \sqrt{\text{area}}$).
- **Minimum feature size (per modeler guidance):** the modeling lead advised that features smaller than ~10 m in every direction may be dropped. We applied a **conservative 8 m** threshold (retaining slightly more detail than the 10 m allowance). Implemented via each polygon's **oriented bounding box**: a polygon is dropped only if **both** of its oriented sides are < 8 m - so thin real features (e.g., a 2 m $\times$ 500 m spit) are **preserved** while true small blobs are removed.

1.5 Processing Pipeline (summary)

All geometry work was performed in PostGIS on a metric CRS (EPSG:6344, UTM 15N) so every distance parameter (5 m buffer, 8 m feature size, etc.) is literal meters; final products are reprojected to WGS84 for delivery.

Source shapefiles (5)

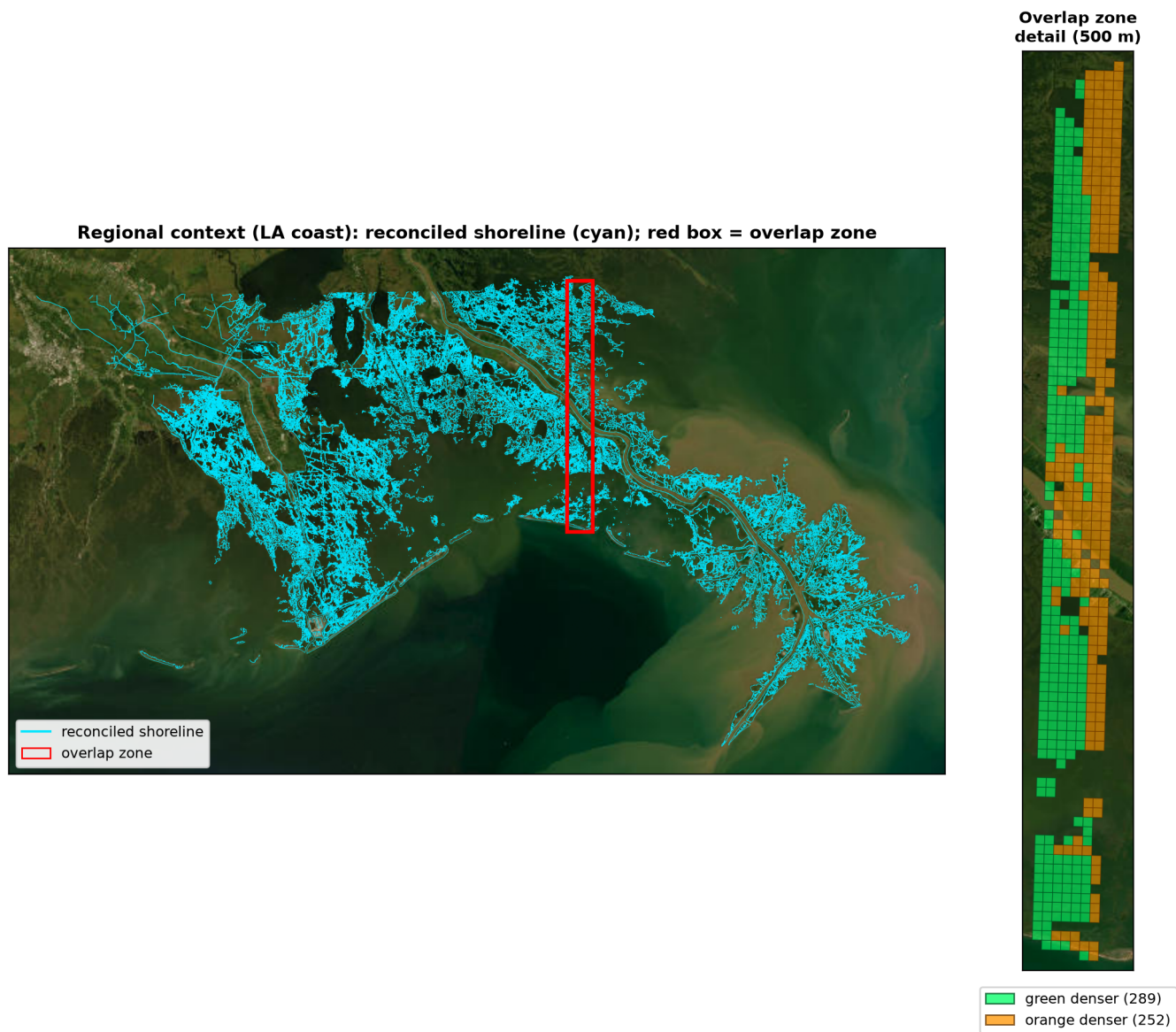
```
| load to PostGIS; keep original geometry + a metric (EPSG:6344) working copy
v
```

```
[Detail assessment] vertices/km per dataset (green 710 ... MS_TX 111)
|
```

```
v
```

```
[Authority map] 500 m cells; per cell winner = denser of green/orange
|
```

```
v
```



Basemap: Esri World Imagery (Esri, Maxar, Earthstar Geographics). Data transformed to Web Mercator (EPSG:3857) for basemap display.

Figure 5: Detail-based authority map over the overlap zone (500 m cells): the denser dataset wins each cell (green 289, orange 252), shown on Esri World Imagery. Left: regional context on the Louisiana coast with the reconciled shoreline (cyan) and the overlap zone (red box). Right: zoomed detail. Data transformed to Web Mercator (EPSG:3857) for basemap display.

```

[Overlap removal]    per cell: keep winner whole; clip loser within 5 m of winner
                     |
                     | (ST_Difference against 5 m buffer) -> removes duplicate co-tracing
                     v
[Fragment cleanup]   drop parts that are BOTH < 15 m AND within 6 m of reference
                     |
                     v
[Assemble network]   detail-reconciled green/orange + untouched purple + red
                     |
                     v
[Polygonize]         ST_Node -> ST_Polygonize (tiled for scale)
                     |
                     v
[Poly cleanup]       remove nodding-artifact slivers; drop features < 8 m in every
                     |                      direction (oriented-bbox; thin spits preserved)
                     v
RC1 deliverables (reproject -> WGS84 / EPSG:4326):
- gom_shoreline_lines_RC1.gpkg    (230,022 lines)
- gom_shoreline_polys_RC1.gpkg    (88,807 polygons)

```

Key parameters (all tunable, all physically meaningful):

Parameter	Value	Meaning
Working CRS	EPSG:6344 (m)	metric so distances are literal
Overlap buffer N	5 m	“within N m = duplicate co-tracing”
Fragment rule	< 15 m AND < 6 m of ref	remove clip debris only
Authority cell size	500 m	resolution of detail-preference decision
Min feature size	8 m (both oriented sides)	conservative vs. modeler’s 10 m

1.6 Validation & Results

Deliverables:

Layer	Features	CRS	Geometry
Merged shoreline (lines)	230,022	WGS84 (4326)	LineString
Companion polygons	88,807	WGS84 (4326)	Polygon (897 km ² total)

Tests performed:

- Overlap eliminated.** Buffer-difference reduced residual overlap between reconciled layers to **0.0 m** (measured), versus the concave-hull method which always left overlap. This directly addresses the failure that motivated the work.
- Tangle resolved.** At a representative complex-marsh location, green/orange crossings dropped from **1,076 to 0** after detail-based reconciliation - a coherent single shoreline replaces the interleaved mat.
- Polygon geometry 100% valid.** All 88,807 polygons pass ST_IsValid; **0 invalid, 0 empty, 0 non-polygon**. No self-intersections or degenerate rings that would trip a mesher.
- Detail preserved.** Full source vertex density retained (no smoothing/thinning); per-zone detail preference keeps the finest available tracing everywhere (green in 289 cells, orange in 252).
- Open-shoreline accounting (honest dangle metric).** ~24% of line length does **not** close into a polygon. This is expected and correct: it comprises genuinely open coastline, the thin western strip (red), unique non-duplicated segments, and real gaps present in the *source* data. These are **preserved in the line layer** rather than closed with fabricated shoreline. The remaining ~76% forms closed polygons.

Fig 5. Polygon product size distribution (88,807 polygons)

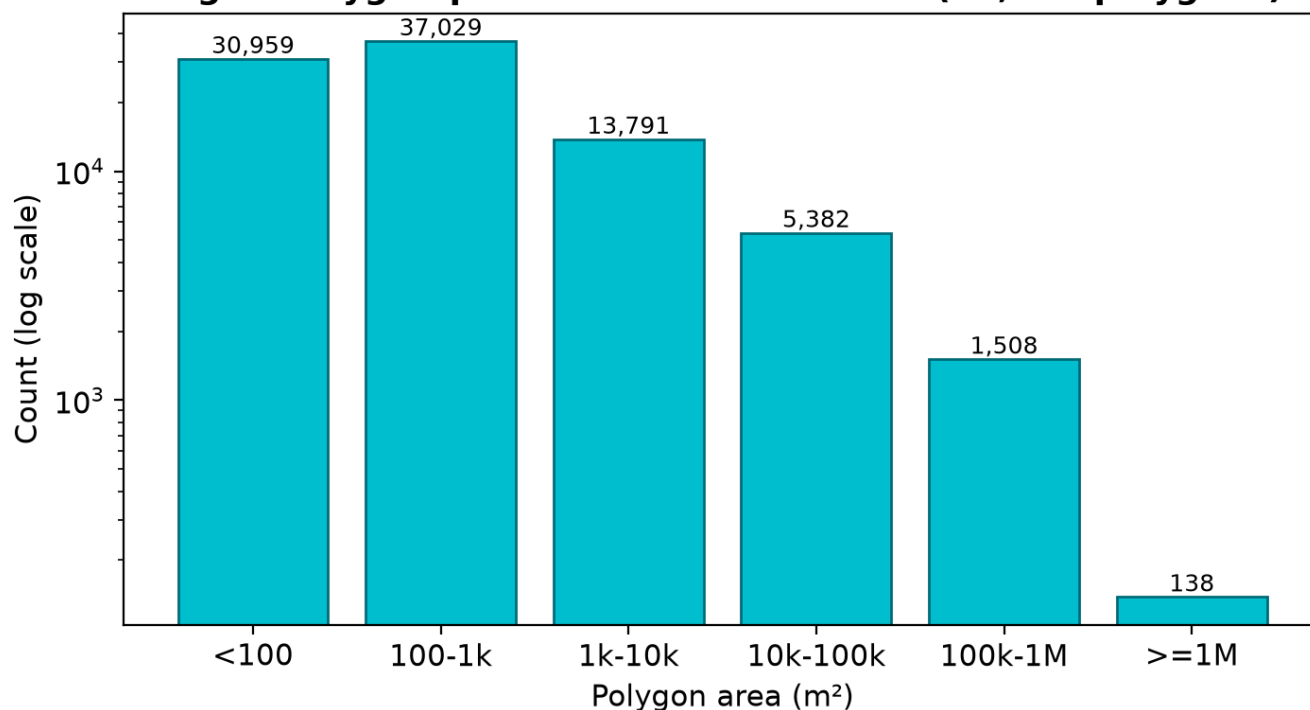


Figure 6: Polygon product size distribution (88,807 polygons; count on log scale).

Assessment: Based on the above, the RC1 products are structurally clean and should ingest into SMS without the overlap/topology problems that broke prior attempts. The polygon layer is watertight where real features close; the line layer preserves everything (including legitimately open features) for use as boundary constraints.

1.7 Optional Polish & Contingencies (if SMS objects)

RC1 was deliberately released without the following optional steps, to obtain real SMS feedback before further processing. **If SMS reports problems, these are ready options:**

- **Cell-boundary snapping.** Reconciliation decisions are made per 500 m cell. Visual review found the cell-to-cell transitions clean, but a *targeted* endpoint-snap along cell boundaries can be applied if SMS finds small gaps/jogs there. (A full whole-network self-snap was tested and found computationally impractical and unnecessary; a scoped boundary-only snap is the efficient alternative.)
- **Global line noding.** For a strictly, topologically fully-connected line network (every touch a shared node), a global node pass can be applied. Deferred because it is expensive and may be unnecessary for SMS's ingestion.
- **Feature-size threshold.** The 8 m minimum is conservative (modeler allowance was 10 m). It can be raised toward 10 m to remove more small features, or lowered to keep more, per modeler preference - a one-parameter change.
- **Additional gap closing.** Some dangles are real source-data gaps (e.g., an island outline broken in the source with gaps up to hundreds of meters). We did **not** fabricate shoreline to close these. If specific features must be closed, they can be addressed in a reviewed, targeted pass.

All of the above are parameter/scope adjustments on the existing pipeline, not redesigns.

1.8 What We Did NOT Do (data integrity statement)

To be explicit for downstream users:

- **No shoreline was smoothed, generalized, or thinned.** Full source vertex detail is retained (this is why the files are large).
- **No shoreline was fabricated.** Gaps present in the source remain open; we did not invent linework to force polygon closure.
- **Source datasets were never modified.** All processing produced new derived layers from read-only originals; every step is reproducible from source.
- **All removals were rule-based and quantified** (duplicate co-tracing within 5 m; clip debris < 15 m AND < 6 m of reference; artifact slivers; features < 8 m in every direction). No manual/hand editing of geometry.

The work is best characterized as **rigorous, rule-based geoprocessing of complex overlapping datasets** - reconciliation and derivation, not manipulation.

1.9 Files

File	Contents
RC1/gom_shoreline_lines_RC1.gpkg	Merged shoreline line layer (WGS84)
RC1/gom_shoreline_polys_RC1.gpkg	Companion polygon layer (WGS84)
gom_shoreline_pipeline.sql	Reproducible PostGIS pipeline (functions + driver)
This document	Methods & results

1.10 Acknowledgments

AI Assistance Disclosure: Portions of this report were prepared with the assistance of an AI language model (Claude Opus 4.8, via OpenCode). The analysis, direction, interpretation, and conclusions are the author’s own; AI was used as a drafting and computational aid under the author’s direction. The author has reviewed all content and takes full responsibility for its accuracy and conclusions.

RC1 - provided for modeler evaluation. Parameters and optional polish steps can be tuned per SMS results before a final release.